DTL ASSIGNMENT 5

GIT AND GITHUB

DIV 1 (S1)

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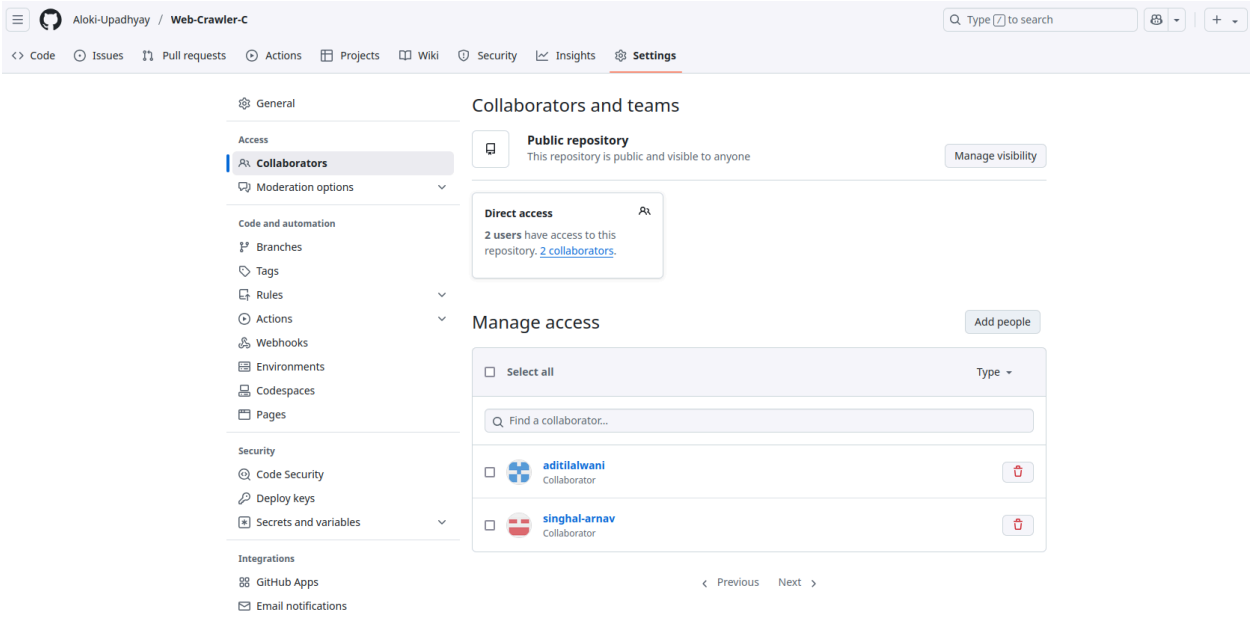
1) Creating a local repository - git init

```
aloki@aloki-HP-Laptop-15s-fq4xxx:~/Web-Crawler-C$ git init
hint: Using 'master' as the name for the initial branch. This default branch nam
e
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/aloki/Web-Crawler-C/.git/
```

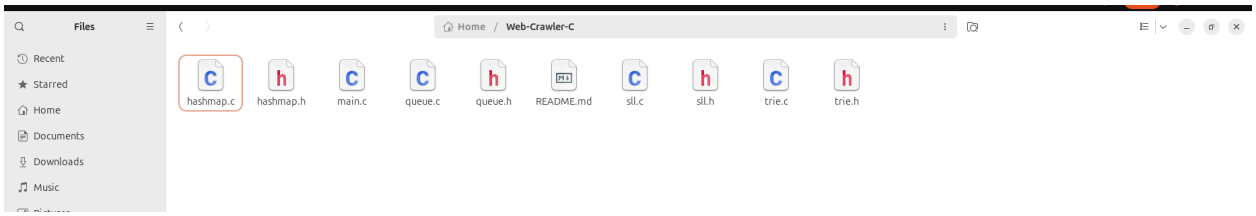
2) Creating a remote repository on GitHub and adding collaborators



3) Adding GitHub repository as remote for local repository - git remote add

```
aloki@aloki-HP-Laptop-15s-fq4xxx:~/Web-Crawler-C$ git remote add origin https://github.com/Aloki-Upadhyay/Web-Crawler-C.git
aloki@aloki-HP-Laptop-15s-fq4xxx:~/Web-Crawler-C$ git add --all
aloki@aloki-HP-Laptop-15s-fq4xxx:~/Web-Crawler-C$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   README.md
        new file:   hashmap.c
        new file:   hashmap.h
        new file:   main.c
        new file:   queue.c
        new file:   queue.h
        new file:   sll.c
        new file:   sll.h
        new file:   trie.c
        new file:   trie.h
```

4) Adding files to empty folder



5) Adding files to staging index and verifying - git add & git status

```
aloki@aloki-HP-Laptop-15s-fq4xxx:~/Web-Crawler-C$ git add --all
aloki@aloki-HP-Laptop-15s-fq4xxx:~/Web-Crawler-C$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   README.md
        new file:   hashmap.c
        new file:   hashmap.h
        new file:   main.c
        new file:   queue.c
        new file:   queue.h
        new file:   sll.c
        new file:   sll.h
        new file:   trie.c
        new file:   trie.h
```

6) Making a commit (in the local repository) - git commit

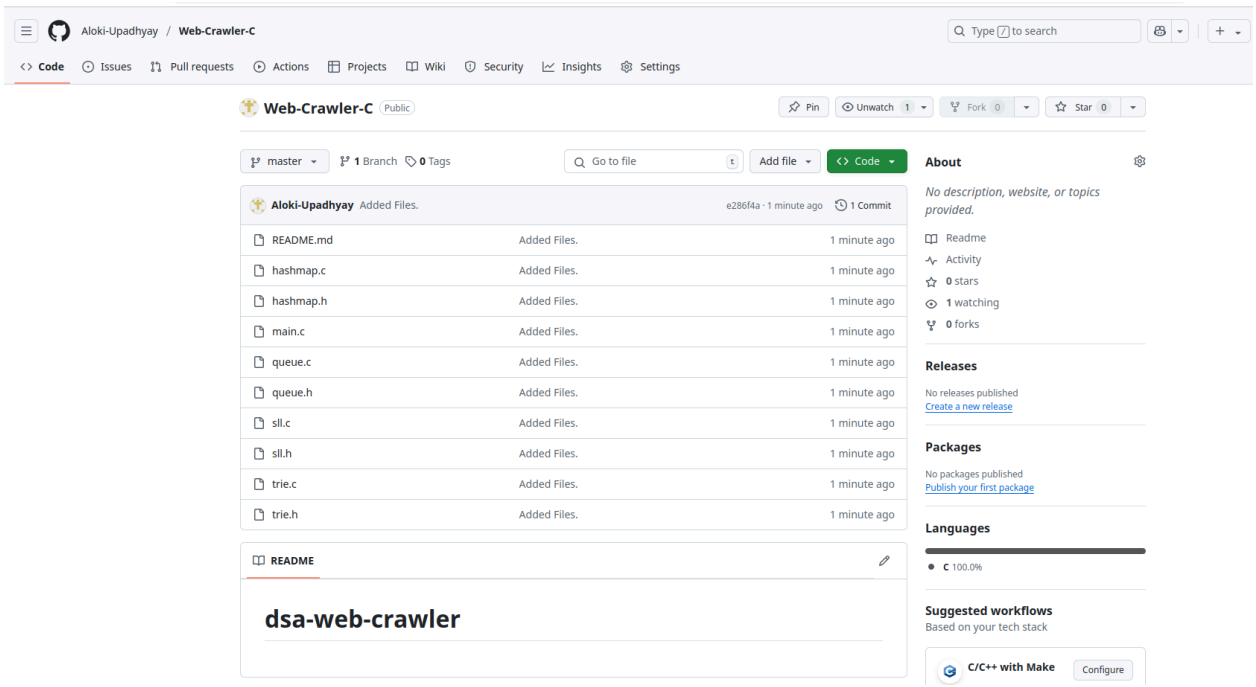
```
aloki@aloki-HP-Laptop-15s-fq4xxx:~/Web-Crawler-C$ git commit -m "Added Files."
[master (root-commit) e286f4a] Added Files.
 10 files changed, 839 insertions(+)
 create mode 100644 README.md
 create mode 100644 hashmap.c
 create mode 100644 hashmap.h
 create mode 100644 main.c
 create mode 100644 queue.c
 create mode 100644 queue.h
 create mode 100644 sll.c
 create mode 100644 sll.h
 create mode 100644 trie.c
 create mode 100644 trie.h
```

7) Pushing changes to remote repository - git push

[Note: -u flag is used here to set the master branch of the remote repository (origin) as the upstream for the master branch of the local repository. Its full form is –set-upstream.]

```
aloki@aloki-HP-Laptop-15s-fq4xxx:~/Web-Crawler-C$ git push -u origin master
Enumerating objects: 12, done.
Counting objects: 100% (12/12), done.
Delta compression using up to 8 threads
Compressing objects: 100% (11/11), done.
Writing objects: 100% (12/12), 9.84 KiB | 1.97 MiB/s, done.
Total 12 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/Aloki-Upadhyay/Web-Crawler-C.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.
```

8) Verifying changes made in GitHub repository



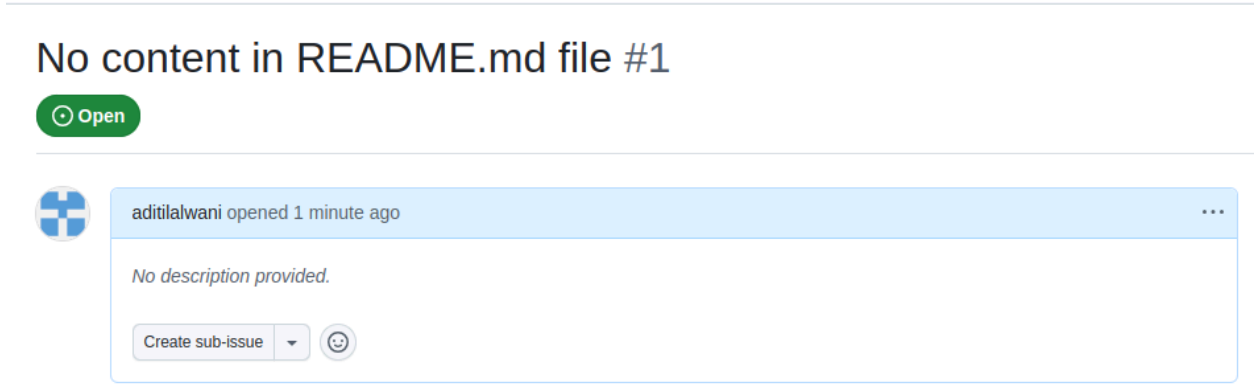
9) GitHub repository cloned by collaborators (Aditi, Arnav) - git clone

```
aditi@aditi-HP-Pavilion-Laptop-14-dv0xxx:~$ git clone https://github.com/Aloki-Upadhyay/Web-Crawler-C.git
Cloning into 'Web-Crawler-C'...
remote: Enumerating objects: 12, done.
remote: Counting objects: 100% (12/12), done.
remote: Compressing objects: 100% (11/11), done.
remote: Total 12 (delta 0), reused 12 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (12/12), 9.84 KiB | 9.84 MiB/s, done.
```

MINGW64:/c/Users/singh

```
singh@Arnav-Shilpi-L1 MINGW64 ~
$ git clone https://github.com/Aloki-Upadhyay/Web-Crawler-C.git
Cloning into 'Web-Crawler-C'...
remote: Enumerating objects: 12, done.
remote: Counting objects: 100% (12/12), done.
remote: Compressing objects: 100% (11/11), done.
remote: Total 12 (delta 0), reused 12 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (12/12), 9.84 KiB | 3.28 MiB/s, done.
```

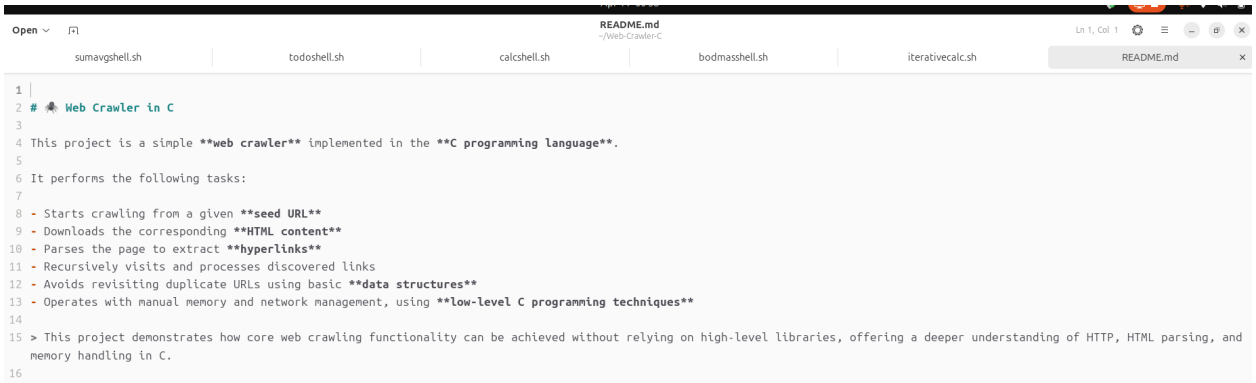
10) Issue created on GitHub (by Aditi)



11) Creating new branch fix-issue-1 in local repository

```
aloki@aloki-HP-Laptop-15s-fq4xxx:~/Web-Crawler-C$ git checkout -b fix-issue-1
Switched to a new branch 'fix-issue-1'
aloki@aloki-HP-Laptop-15s-fq4xxx:~/Web-Crawler-C$ git branch
* fix-issue-1
  master
aloki@aloki-HP-Laptop-15s-fq4xxx:~/Web-Crawler-C$
```

12) Changes made - Content added to README.md file



13) Comparing with previous commit - git diff

```
aloki@aloki-HP-Laptop-15s-fq4xxx:~/Web-Crawler-C$ git diff
diff --git a/README.md b/README.md
index aa8903e..de22301 100644
--- a/README.md
+++ b/README.md
@@ -1 +1,16 @@
-# dsa-web-crawler
\ No newline at end of file
+
+# 🕸Web Crawler in C
+
+This project is a simple **web crawler** implemented in the **C programming language**.
+
+It performs the following tasks:
+
+- Starts crawling from a given **seed URL**
+- Downloads the corresponding **HTML content**
+- Parses the page to extract **hyperlinks**
+- Recursively visits and processes discovered links
+- Avoids revisiting duplicate URLs using basic **data structures**
+- Operates with manual memory and network management, using **low-level C programming techniques**
+
+> This project demonstrates how core web crawling functionality can be achieved without relying on high-level libraries, offering
a deeper understanding of HTTP, HTML parsing, and memory handling in C.
+
```

14) Adding file to staging index and making a commit

```
+
aloki@aloki-HP-Laptop-15s-fq4xxx:~/Web-Crawler-C$ git add README.md
```

```
aloki@aloki-HP-Laptop-15s-fq4xxx:~/Web-Crawler-C$ git commit -m "Fixes #1: description added to README.md file"
[fix-issue-1 f3a4e4c] Fixes #1: description added to README.md file
 1 file changed, 16 insertions(+), 1 deletion(-)
```

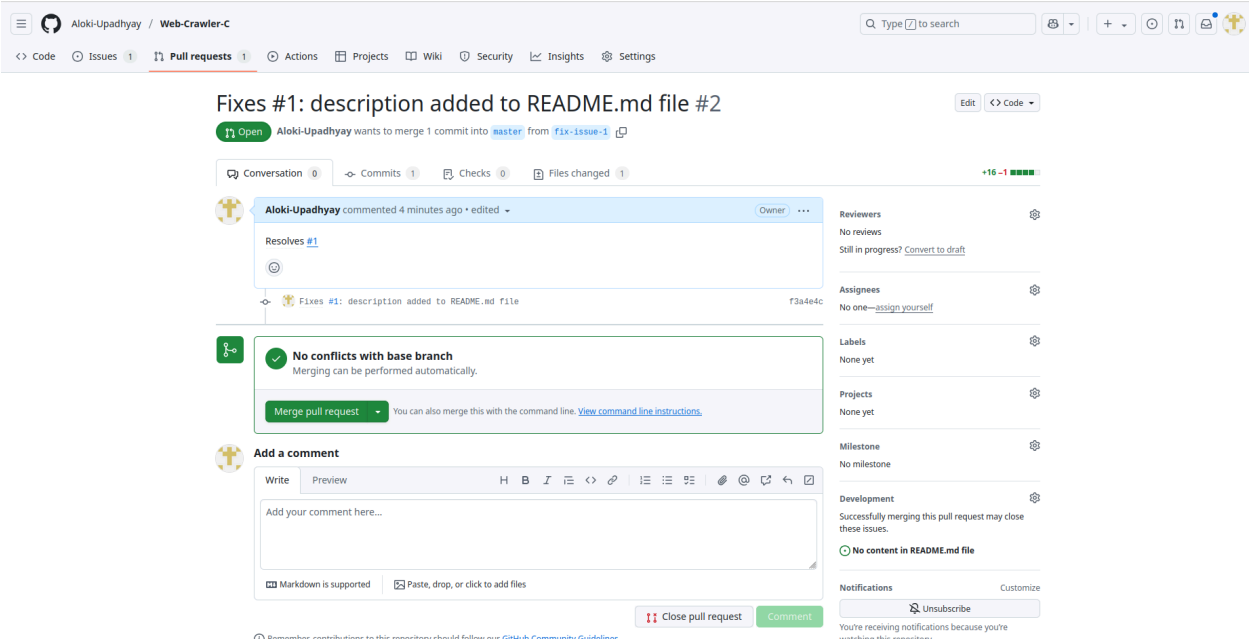
15) Viewing list of commits - git log

```
aloki@aloki-HP-Laptop-15s-fq4xxx:~/Web-Crawler-C$ git log
commit f3a4e4cc662e0eaa23605e711cb9be9a97ae9b64 (HEAD -> fix-issue-1)
Author: Aloki-Upadhyay <aloki.upadhyay@gmail.com>
Date:   Fri Apr 11 01:02:19 2025 +0530

    Fixes #1: description added to README.md file

commit e286f4a2ac50723914d82492e6f300a99c8f7783 (origin/master, master)
Author: Aloki-Upadhyay <aloki.upadhyay@gmail.com>
Date:   Fri Apr 11 00:44:12 2025 +0530

    Added Files.
```

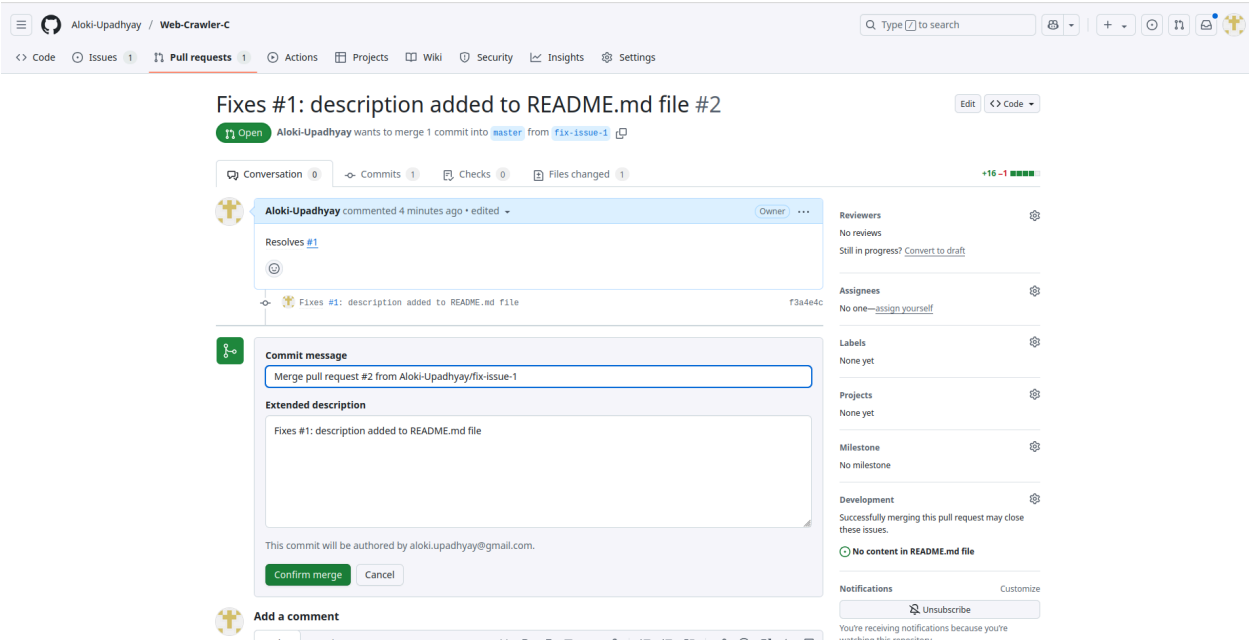
16) Pushing changes to new branch fix-issue-1 in remote repository

```
aloki@aloki-HP-Laptop-15s-fq4xxx:~/Web-Crawler-C$ git push -u origin fix-issue-1
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 742 bytes | 742.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'fix-issue-1' on GitHub by visiting:
remote:      https://github.com/Aloki-Upadhyay/Web-Crawler-C/pull/new/fix-issue-1
remote:
To https://github.com/Aloki-Upadhyay/Web-Crawler-C.git
 * [new branch]      fix-issue-1 -> fix-issue-1
branch 'fix-issue-1' set up to track 'origin/fix-issue-1'.
```

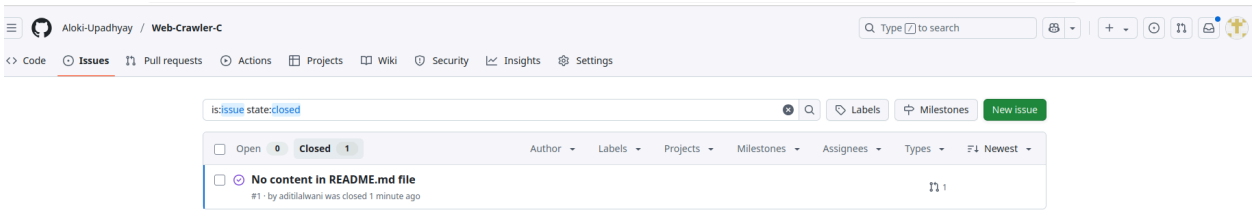
17) Issuing a pull request on GitHub



18) Merging pull request into master branch



19) Issuing closed automatically



20) Downloading and merging changes from remote repository's master branch into local repository's master branch

a) git pull (Aloki)

```
aloki@aloki-HP-Laptop-15s-fq4xxx:~/Web-Crawler-C$ git checkout master
Switched to branch 'master'
Your branch is up to date with 'origin/master'.
aloki@aloki-HP-Laptop-15s-fq4xxx:~/Web-Crawler-C$ git pull
remote: Enumerating objects: 1, done.
remote: Counting objects: 100% (1/1), done.
remote: Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (1/1), 919 bytes | 919.00 KiB/s, done.
From https://github.com/Aloki-Upadhyay/Web-Crawler-C
   e286f4a..b26eb12  master     -> origin/master
Updating e286f4a..b26eb12
Fast-forward
 README.md | 17 ++++++++++++++++-
 1 file changed, 16 insertions(+), 1 deletion(-)
```

b) git fetch & git merge (Arnav, Aditi)

MINGW64:/c/Users/singh/Web-Crawler-C

```
singh@Arnav-Shilpi-L1 MINGW64 ~/Web-Crawler-C (master)
$ git fetch
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 4 (delta 2), reused 2 (delta 1), pack-reused 0 (from 0)
Unpacking objects: 100% (4/4), 1.45 KiB | 78.00 KiB/s, done.
From https://github.com/Aloki-Upadhyay/Web-Crawler-C
   e286f4a..b26eb12  master      -> origin/master
 * [new branch]      fix-issue-1 -> origin/fix-issue-1

singh@Arnav-Shilpi-L1 MINGW64 ~/Web-Crawler-C (master)
$ git merge
Updating e286f4a..b26eb12
Fast-forward
 README.md | 17 +++++++++++++++++-
 1 file changed, 16 insertions(+), 1 deletion(-)

singh@Arnav-Shilpi-L1 MINGW64 ~/Web-Crawler-C (master)
$
```

```
aditi@aditi-HP-Pavilion-Laptop-14-dv0xxx:~/Web-Crawler-C$ git fetch
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 4 (delta 2), reused 2 (delta 1), pack-reused 0 (from 0)
Unpacking objects: 100% (4/4), 1.45 KiB | 745.00 KiB/s, done.
From https://github.com/Aloki-Upadhyay/Web-Crawler-C
   e286f4a..b26eb12  master      -> origin/master
 * [new branch]      fix-issue-1 -> origin/fix-issue-1
aditi@aditi-HP-Pavilion-Laptop-14-dv0xxx:~/Web-Crawler-C$ git merge
Updating e286f4a..b26eb12
Fast-forward
 README.md | 17 +++++++++++++++++-
 1 file changed, 16 insertions(+), 1 deletion(-)
aditi@aditi-HP-Pavilion-Laptop-14-dv0xxx:~/Web-Crawler-C$
```